

# A Comparative Study of Dijkstra’s Algorithm and the Graph $p$ -eikonal Method: Fidelity, Robustness, and Parameter Selection

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## Abstract

This thesis compares Dijkstra’s algorithm and the graph-based  $p$ -eikonal method from the perspectives of path geometry, fidelity, robustness, and practical performance. Experiments on synthetic datasets show that the  $p$ -eikonal method can produce smoother and more stable distance propagations than Dijkstra’s algorithm under weighted perturbations, while still remaining close to the clean reference in appropriate parameter ranges. A fidelity–robustness analysis further suggests that there is no universally optimal value of  $p$ ; instead, small or moderate values often provide the best balance. Additional tests on MNIST confirm that the  $p$ -eikonal method also outperforms Dijkstra’s algorithm in a graph-based classification setting.

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# 1 Introduction

## 1.1 Background

Given a graph of non-negative edges, a source vertex and a target vertex, the shortest path problem seeks an optimal path of the minimal cost. It has long been one of the fundamental problems in plenty of practical fields, including but not limited to operations research, logistics planning and communication routing. In reality, cost can represent distance, duration, energy cost, risk or a composite measure that balances multiple factors.

The shortest path problem is no longer merely an isolated topic in traditional graph theory, but is also gradually becoming an important factor in machine learning. First, in graph learning, manifold learning and semi-supervised learning, distances or costs on graphs have been used to represent geometric relationships among data points. Moreover, in the area of geometric modeling in high-dimension data, where the contrast between sparse and dense regions becomes more significant, researchers are also concerned with whether an ideal distance function should favor strictly discrete shortest paths or a smoothed propagation process that is less sensitive to noise.

In discrete graphs with non-negative edges, Dijkstra’s algorithm and its variants are among the most classical and basic methods to solve the shortest path problem. Their key idea is to expand the known set of optimal distances step by step, and ultimately obtain the shortest distance from the source vertex to all vertices. Due to their clear logic, simple implementation and high computational efficiency, they hold significant positions in both theoretical research and industrial applications.

However, Dijkstra-type methods also have significant limitations. Since Dijkstra-type methods seek the shortest path by combinatorially comparing discrete edge weights, when the edge weights are corrupted by noise like weight noise or abnormal edges, the shortest

path they find may undergo abrupt changes: a small local change can ultimately lead to the entire path turning into another completely different path. These phenomena show that Dijkstra-type methods perform well in finding the path with minimal cost, but not in finding a robust path. This sensitivity to noise becomes even more significant in machine learning scenarios, especially when data graphs contain sampling noise and outliers.

The  $p$ -eikonal equation provides a framework that, while preserving the core features of the shortest path approach, is more stable against noise and perturbations. Unlike Dijkstra-type methods, the  $p$ -eikonal method emphasizes constructing a distance function in a global propagation context. Specifically, the introduction of the parameter  $p$  allows the model to adjust between "discrete paths" and "smooth and robust propagation". However, how  $p$  substantially affects the solution from the perspectives of path shape, sensitivity to noise, and overall numerical performance, remains insignificant. Is its effect continuous and asymptotic, or is there a clear boundary between small and large values of  $p$ ? These questions need to be answered through systematic experiments.

## 1.2 Research Questions

Based on the background above, this thesis mainly focuses on the following three research questions.

### 1.2.1 How does $p$ affect fidelity?

Since the problem under consideration is still that of finding a path of minimal cost, this thesis takes cost as the primary criterion for evaluating fidelity. By definition, Dijkstra's algorithm computes a path with minimal discrete cost. Therefore, the paths obtained by Dijkstra's algorithm in the clean setting, that is, in the absence of noise, are taken as the reference paths. Fidelity is then defined by the extent to which the paths produced by

the  $p$ -eikonal method deviate from those reference paths in terms of cost. In other words, this thesis asks the following question: as  $p$  varies, do the path solutions generated by the  $p$ -eikonal method become closer to the traditional shortest paths, or do they move farther away, under the same clean setting?

### **1.2.2 How does $p$ affect robustness?**

Here, robustness refers to whether the path produced for a fixed  $p$  remains stable under different levels of weighted noise. Accordingly, this thesis evaluates the robustness of  $p$  by repeatedly testing each  $p$  under noisy settings and examining how much the resulting paths deviate from their own reference paths obtained in the corresponding clean setting.

### **1.2.3 Is there a practically usable rule or a reasonable range for selecting $p$ ?**

The third question of this thesis is not simply what the optimal value of  $p$  is, but rather a more pragmatic one: whether there exists a practical principle for choosing  $p$ , or at least a reasonable range of values. Ideally, we hope to identify a single optimal  $p$ . However, if the experimental results do not support a clear and generalizable selection rule, then the goal is to determine a practical range of  $p$  where both fidelity and robustness remain satisfactory.

## **1.3 Main Contributions**

This thesis presents a comparative study of Dijkstra’s algorithm and the graph  $p$ -eikonal method from the perspectives of path geometry, fidelity, robustness, and practical performance on both synthetic and real datasets. The main purpose is not only to compare two methods for shortest-path-type propagation, but also to clarify how the parameter  $p$  affects the behavior of the graph  $p$ -eikonal method in practice.

The experiments lead to several main findings. First, the graph  $p$ -eikonal method and

Dijkstra’s algorithm are closely related but not equivalent. On clean datasets, both methods often produce paths that are geometrically similar, but the extent of similarity depends on the dataset. In particular, the effect of  $p$  is strongest on the Two Moons dataset, more moderate on Obstacle Grid, and relatively weak on Gaussian Mixture.

Second, under weighted perturbations, the graph  $p$ -eikonal method is generally more robust than Dijkstra’s algorithm. Across the synthetic datasets considered in this thesis, Dijkstra’s method tends to show larger path deviations once noise is introduced, whereas the  $p$ -eikonal solutions usually remain more stable. This confirms that the  $p$ -eikonal framework is not merely another way of computing shortest paths, but a more flexible distance-propagation model with improved stability under perturbation.

Third, the experiments do not support the existence of a single universally optimal value of  $p$ . For some datasets, the clean fidelity metric is somewhat sensitive to the choice of  $p$ , and in such cases the fidelity–robustness trade-off helps identify a reasonable operating region. For other datasets, however, many nearby  $p$ -values perform almost equally well. Therefore, the thesis concludes that parameter selection should be interpreted as identifying a stable and practically effective range, rather than a unique best value.

Finally, the MNIST classification experiment shows that the same general conclusion continues to hold in a higher-dimensional graph-based learning setting. The graph  $p$ -eikonal method substantially outperforms the Dijkstra baseline in both accuracy and macro-F1, while the performance differences among the tested  $p$ -values remain limited. This suggests that the practical advantage of the method lies less in tuning an exact optimum and more in providing a robust family of distance-propagation models that perform well across a broad parameter range.

Taken together, these results form the main contribution of this thesis: a unified experimental comparison of Dijkstra-type and graph  $p$ -eikonal methods that highlights the

role of  $p$  in controlling the balance between shortest-path fidelity and robustness, and that shows the practical value of the graph  $p$ -eikonal method in both geometric path analysis and graph-based classification.

## 2 Mathematical Background and Related Works

### 2.1 Graph

#### 2.1.1 Weighted Graph

In this thesis, a weighted graph is defined as

$$G = (X, W),$$

where  $X$  is the set of vertices,

$$X = \{x_1, x_2, \dots, x_n\} \subset \mathbb{R}^d, \quad d \in \mathbb{N},$$

and  $W$  is the matrix of nonnegative edge weights,

$$W = (w_{ij})_{i,j=1}^n, \quad w_{ij} \geq 0.$$

Each weight  $w_{ij}$  represents the similarity between the two vertices  $x_i$  and  $x_j$ . A larger weight indicates that the two vertices are more similar, or equivalently, closer to each other in the graph structure. If  $w_{ij} \approx 0$ , then the two vertices are regarded as far apart. In particular, when  $w_{ij} = 0$ , it is regarded as being no edge from  $x_i$  to  $x_j$ . In general, the weight matrix  $W$  need not be symmetric.

Let  $\mathcal{F}(X)$  denote the space of real-valued functions on  $X$ . For  $u \in \mathcal{F}(X)$  and a vertex

$x_i \in X$ , the graph gradient of  $u$  at  $x_i$  is defined by

$$\nabla_X u(x_i) = (u(x_i) - u(x_1), \dots, u(x_i) - u(x_n)), \quad \nabla_X^j u(x_i) = u(x_i) - u(x_j).$$

### 2.1.2 The Graph Eikonal Equation and Dijkstra's algorithm

In the shortest path problem and its generalized models, it is necessary to specify a set of boundary vertices, which here represents the set of terminal nodes. The thesis denotes this boundary set by  $\partial X \subset X$ . Since  $u$  represents the distance to the boundary vertices, it is natural to impose the boundary condition

$$u(x) = 0, \quad x \in \partial X.$$

On the graph, the thesis defines the edge cost as the reciprocal of the edge weight:

$$e_{ij} = w_{ij}^{-1}, \quad 1 \leq i, j \leq n.$$

Then the graph distance between two vertices  $x_i$  and  $x_j$  is defined as

$$d_G(x_i, x_j) = \min_{m \geq 1} \min_{\gamma \in I_n^m} \left( w_{i, \gamma_1}^{-1} + \sum_{k=1}^{m-1} w_{\gamma_k, \gamma_{k+1}}^{-1} + w_{\gamma_m, j}^{-1} \right),$$

where  $I_n = \{1, 2, \dots, n\}$ ,  $\gamma = (\gamma_1, \dots, \gamma_m)$ , and

$$\{x_{\gamma_1}, \dots, x_{\gamma_m}\} \subset X.$$

Now define  $u$  as the distance function to the boundary set:

$$u(x_i) = d_G(x_i, \partial X) = \min_{x_j \in \partial X} d_G(x_i, x_j).$$

For  $x_i \in X \setminus \partial X$ , this satisfies the Bellman relation

$$u(x_i) = \min_{x_j \in X} (u(x_j) + w_{ij}^{-1}).$$

This is precisely the dynamic programming relation underlying Dijkstra's algorithm, in the classical sense described in *Introduction to Algorithms* by Cormen, Leiserson, Rivest, and Stein: the optimal cost from vertex  $x_i$  to the boundary is equal to the minimum, over all neighboring vertices, of the cost of moving first to that vertex and then following an optimal path from there to the boundary.

Equivalently, the Bellman relation can be rewritten as

$$\begin{cases} \max_{x_j \in X} w_{ij} (u(x_i) - u(x_j)) = 1, & x_i \in X \setminus \partial X. \\ u(x_i) = 0, & x_i \in \partial X \end{cases}$$

*Proof.* From

$$u(x_i) = \min_{x_j \in X} (u(x_j) + w_{ij}^{-1}),$$

obtain

$$u(x_i) - u(x_j) - w_{ij}^{-1} \leq 0, \quad \forall x_j \in X.$$

Since the minimum is attained for at least one  $x_j$ , there exists an index  $j$  such that equality holds. Hence

$$\max_{x_j \in X} (u(x_i) - u(x_j) - w_{ij}^{-1}) = 0.$$

Multiplying by  $w_{ij} \geq 0$  yields

$$\max_{x_j \in X} w_{ij} (u(x_i) - u(x_j)) = 1.$$

□

Next, suppose a positive function  $f$  on the vertices is introduced to represent a local propagation cost. Then the one-step propagation cost from  $x_i$  to  $x_j$  becomes  $f(x_i)w_{ij}^{-1}$ , and the distance-to-boundary relation is modified to

$$u(x_i) = \min_{x_j \in X} (u(x_j) + w_{ij}^{-1} f(x_i)), \quad x_i \in X \setminus \partial X.$$

Equivalently, the relation can be rewritten as the graph eikonal equation

$$\begin{cases} \max_{x_j \in X} w_{ij} (u(x_i) - u(x_j)) = f(x_i), & x_i \in X \setminus \partial X. \\ u(x_i) = 0, & x_i \in \partial X \end{cases}$$

In machine learning, a common choice of  $f$  is

$$f(x_i) = \hat{\rho}(x_i)^{-\alpha}, \quad \alpha \geq 0,$$

where  $\hat{\rho}$  is a kernel density estimator. Under this choice, the propagation cost is lower in high-density regions and higher in sparse regions. As a result, the induced paths are encouraged to pass through high-density areas rather than sparse ones.

## 2.2 Mathematical background of the graph $p$ -eikonal method

This sub chapter primarily outlines the mathematical background of the  $p$ -eikonal equation, including definition, well-posedness, and robustness analysis of the  $p$ -eikonal method, mainly based on and adapted from Calder and Ettehad (2022, Section 2).

From the graph eikonal equation, it can be seen that the Dijkstra-type shortest-path method essentially relies on only "the best neighbor" at each vertex. This also reflects its combinatorial optimization nature: the direction of propagation is determined by the locally optimal edge.

However, this reliance on the locally optimal edge also introduces significant robustness issues. Adding an unreasonable increase in the weight of an edge can cause the optimal neighbor of the current node to switch instantaneously, resulting in a drastic change in the shortest path and its corresponding distance function.

To mitigate this over-reliance on a single neighbor, Calder and Ettehad introduced the graph  $p$ -eikonal operator. Its basic idea is to no longer focus solely on the optimal neighbor, but rather allow all "upwind neighbors" to participate in determining the node's value. Here, "upwind neighbors" refer to those that are closer to the boundary than  $x_i$ :

$$\{x_j \in X, u(x_j) < u(x_i)\}$$

Therefore, Calder and Ettehad introduced an operator

$$A_{G_p} u(x_i) = \sum_{j=1}^n w_{ij} (u(x_i) - u(x_j))_+^p = f(x_i), \quad x_i \in X \setminus \partial X$$

where

$$(a)_+ = \max\{a, 0\}$$

Therefore, the graph  $p$ -eikonal equation is obtained:

$$\begin{cases} A_{G_p} u(x_i) = \sum_{j=1}^n w_{ij} (u(x_i) - u(x_j))_+^p = f(x_i), & x_i \in X \setminus \partial X. \\ u(x_i) = 0, & x_i \in \partial X \end{cases}$$

In Calder and Ettehad's thesis, several theoretical properties of the graph  $p$ -eikonal equation are rigorously established, which justify viewing it as a generalized shortest path model:

The following results, adapted from Calder and Ettehad (2022), summarize several basic theoretical properties of the graph  $p$ -eikonal equation.

**Theorem** (Existence and Uniqueness). Suppose  $p > 0$ ,  $f > 0$ , and the graph is connected. Then the graph  $p$ -eikonal equation admits a unique solution  $u \in \mathcal{F}(X)$ .

**Theorem** (Comparability with Graph Distance). Define

$$K = \max_{1 \leq i \leq n} \sum_{j=1}^n \mathbf{1}_{\{w_{ji} > 0\}}.$$

Then, for every  $x_i \in X$ ,

$$K^{-1/p} \left( \min_X f^{1/p} \right) d_{G^{1/p}}(x_i, \partial X) \leq u(x_i) \leq \left( \max_X f^{1/p} \right) d_{G^{1/p}}(x_i, \partial X).$$

Here,  $d_{G^{1/p}}$  denotes the graph distance associated with edge cost  $w_{ij}^{-1/p}$ , namely

$$d_{G^{1/p}}(x_i, \partial X) = \min_{x_j \in \partial X} d_{G^{1/p}}(x_i, x_j),$$

where

$$d_{G^{1/p}}(x_i, x_j) = \min_{m \geq 1} \min_{\gamma \in I_n^m} \left( w_{i, \gamma_1}^{-1/p} + \sum_{k=1}^{m-1} w_{\gamma_k, \gamma_{k+1}}^{-1/p} + w_{\gamma_m, j}^{-1/p} \right).$$

**Theorem** (Robustness with Respect to Nonnegative Perturbations). Suppose that all entries

of the perturbation matrix  $\delta W$  are nonnegative, and define

$$\widetilde{W} = W + \delta W.$$

Let  $G = (X, W)$ ,  $\widetilde{G} = (X, \widetilde{W})$ , and  $\delta G = (X, \delta W)$ . Let  $u, \tilde{u} \in \mathcal{F}(X)$  be the respective solutions of

$$\begin{cases} A_{G,p}u(x_i) = f(x_i), & x_i \in X \setminus \partial X, \\ u(x_i) = 0, & x_i \in \partial X, \end{cases} \quad \begin{cases} A_{\widetilde{G},p}\tilde{u}(x_i) = f(x_i), & x_i \in X \setminus \partial X, \\ \tilde{u}(x_i) = 0, & x_i \in \partial X. \end{cases}$$

Then, for all  $x_i \in X$ ,

$$0 \leq \frac{u(x_i) - \tilde{u}(x_i)}{u(x_i)} \leq \left( \max_{X \setminus \partial X} \frac{A_{\delta G,p}u}{f} \right)^{1/p}.$$

These results show that the graph  $p$ -eikonal equation retains several key characteristics expected of a distance-based propagation model: it is well-posed, remains quantitatively comparable to an appropriate graph distance, and admits an explicit robustness estimate under nonnegative perturbations of the edge weights.

From the perspective of parameter interpretation,  $p$  controls how information from neighboring vertices is aggregated. When  $p$  is small, contributions from a broader set of neighbors are incorporated into the update. When  $p$  is large, the largest upwind differences become increasingly dominant. Accordingly, Calder and Ettehad state in the abstract and in Section 2.3 that  $p = 1$  is the most robust choice with respect to graph perturbations, whereas as  $p \rightarrow \infty$ , the model gradually approaches the behavior of the classical shortest path distance.

## 3 Experiments

### 3.1 Datasets

To compare the performance of Dijkstra’s algorithm and the graph  $p$ -eikonal method under different geometric structures, this thesis considers three representative datasets: Two Moons, Obstacle Grid, and Gaussian Mixture. These datasets reflect different geometric features and application scenarios, and therefore provide a broad basis for comparing the two methods in terms of path shape, shortest-path preference, and robustness to noise.

#### (1) Two Moons

The Two Moons dataset is a standard example of a non-convex manifold. It consists of two crescent-shaped manifolds that lie close to each other without directly overlapping. This dataset is representative because, although some points may be close in Euclidean distance, a reasonable path should follow the manifold structure rather than cross the geometric gap directly.

#### (2) Obstacle Grid

The Obstacle Grid dataset is a two-dimensional regular grid with obstacle regions removed or assigned prohibitively high traversal cost. It represents a classical path-planning problem: a feasible path from a source on the left to a target on the right must avoid intermediate obstacles. This dataset allows a direct comparison of the propagation patterns, obstacle-avoidance behavior, and path preferences of different methods near barriers.

#### (3) Gaussian Mixture

The Gaussian Mixture dataset consists of two Gaussian clusters and represents a typical clustered data structure. It is suitable for examining how different path-based methods

behave on data with nonuniform density and separated groups.

### 3.2 Graph Construction

For each dataset, this thesis first constructs a  $k$ -nearest neighbor graph from the sampled points. More precisely, given a dataset  $X$ , the  $k$  nearest neighbors of each vertex are identified, and weighted edges are added accordingly to form the weight matrix  $W$ . To ensure that a feasible path exists between the source and the target, the thesis does not fix a single value of  $k$  in advance. Instead,  $k$  is increased progressively over a candidate set until a connected graph is obtained.

To avoid additional uncertainty arising from multiple choices of source and target points, this thesis adopts a strict single-source, single-target setting. A representative source point is selected from the candidate source region, and a representative target point is selected from the candidate target region, so that each dataset yields one clearly defined path-planning task. This makes the path comparison between different methods more direct and interpretable.

### 3.3 Settings

To investigate the robustness of Dijkstra’s algorithm and the graph  $p$ -eikonal method under perturbations, this thesis uses weighted noise for all datasets. Specifically, let  $W$  denote the edge-weight matrix of the original graph. The noisy weight matrix  $W^{(\delta)}$  is defined by

$$W_{ij}^{(\delta)} = W_{ij}(1 + \delta\xi_{ij}),$$

where  $\xi_{ij}$  is a random variable drawn from the standard normal distribution  $\mathcal{N}(0, 1)$ , and  $\delta$  controls the noise intensity.

To avoid introducing additional density-dependent effects into the model and ensure a fair comparison between the graph  $p$ -eikonal method and Dijkstra’s algorithm, in all experiments, the source term is chosen as

$$f \equiv 1.$$

To systematically study how the behavior of the graph  $p$ -eikonal method changes with the parameter  $p$ , this thesis considers a range of  $p$ -values from small to large:

$$p \in \{1, 2, 4, 8, 16, 32, 64, 128\}.$$

### 3.4 Metric

To systematically compare the behavior of the  $p$ -eikonal method under different  $p$ , this thesis aims to study how faithful and robust the solutions of different values of  $p$  are under clean and noisy settings.

#### 3.4.1 Fidelity Metric

Fidelity measures how much a path extracted by a certain  $p$ -eikonal method deviates from the shortest path derived by Dijkstra’s method in terms of cost.

It is important to note that this thesis does not use ”geometric distance to a clean Dijkstra path” as the primary definition of fidelity. This is because, for some datasets, especially those with multiple solutions, Dijkstra’s algorithm may only return one of many paths with similar costs. Treating this single path as the only reference would misclassify geometrically different but nearly cost-equivalent paths as having low fidelity.

Suppose that, under the clean setting, the reference path extracted by the Dijkstra

method is  $\gamma_{\text{Dijkstra}}^{\text{clean}}$ , and its total cost under the clean graph is

$$C^* = C_{\text{clean}}(\gamma_{\text{Dijkstra}}^{\text{clean}})$$

Then the fidelity loss is defined as

$$\text{FidLoss}(p) = \frac{|C_{\text{clean}}(\gamma_p) - C^*|}{C^*}$$

The advantage of this definition is that, as long as a path remains close to optimal under the clean objective, it will not be overly penalized even if its geometry differs from the reference path.

To facilitate subsequent comparison with robustness on the same graph, this thesis further transforms the fidelity loss into a fidelity score for which higher is better:

$$\text{Fidelity}(p) = \frac{1}{1 + \text{FidLoss}(p)}$$

### 3.4.2 Path Similarity

This thesis uses robustness to characterize whether, when the graph is disturbed by noise, the extracted paths of a fixed  $p$  remain stable, with regard to the same  $p$ 's own reference path under the clean setting.

To select a metric to evaluate path similarity, this thesis primarily references two thesiss.

Chang et al. provide a systematic review of trajectory similarity measures, discussing discrete Fréchet, Hausdorff, DTW, ERP, and other methods. This thesis aims to compare the similarity of the entire path from source to target, not just the proximity of the point sets traversed by the two paths. Therefore, simply looking at the proximity of sets is not ideal, and the thesis needs to take ordering into consideration. Eiter and Mannila pointed

out that the advantage of Fréchet distance is that it considers not only location but also the order of points on the curve, making it often more suitable than Hausdorff distance when comparing curves.

The discrete Fréchet distance, mainly based on Eiter and Mannila's thesis, is outlined below.

Let two discrete paths be given by

$$P = (u_1, u_2, \dots, u_p), \quad Q = (v_1, v_2, \dots, v_q).$$

The key idea of the discrete Fréchet distance is not to match the two sets of points arbitrarily, but to construct an *ordered coupling* between the two paths. A coupling is a sequence of matched pairs

$$(u_{a_1}, v_{b_1}), (u_{a_2}, v_{b_2}), \dots, (u_{a_m}, v_{b_m}),$$

satisfying the following conditions:

- it starts at  $(u_1, v_1)$  and ends at  $(u_p, v_q)$ ;
- at each step, one may move forward along the first path, or along the second path, or along both paths simultaneously;
- backward movement is not allowed.

Equivalently, the index sequences  $(a_k)$  and  $(b_k)$  must both be nondecreasing. This reflects the essential requirement that the ordering of points along each path must be preserved.

For any coupling  $L$ , its cost is defined as the maximum distance among all matched pairs in the coupling:

$$\|L\| = \max_{k=1, \dots, m} d(u_{a_k}, v_{b_k}),$$

The discrete Fréchet distance is then defined as the minimum possible value of this maximum over all admissible couplings:

$$d_F^{(d)}(P, Q) = \min_L \|L\|.$$

Intuitively, this is equivalent to the classical “man walking a dog” interpretation: a person walks along one path and a dog walks along the other; both may adjust their speeds, but neither is allowed to move backward. The discrete Fréchet distance is the length of the shortest leash required for both to traverse their entire paths. In this way, the definition captures both geometric closeness and the sequential ordering of the paths.

### 3.4.3 Robustness Metric

For a fixed  $p$ , denote a path obtained under the clean setting as  $\gamma_p^{\text{clean}}$ , and a path obtained under noise of extent  $\epsilon$  as  $\gamma_p^{\text{noisy}, \epsilon}$ . Then a natural metric to evaluate this deviation is

$$\Delta_p(\epsilon) = \frac{d_F(\gamma_p^{\text{clean}}, \gamma_p^{\text{noisy}, \epsilon})}{\text{Len}(\gamma_p^{\text{clean}})}$$

However, since there may be more than one reference path in the clean setting, if we still insist on choosing only one path as the sole benchmark, then even if a noisy path is very close to another clean reference, it may be misjudged as “unstable.” Therefore, a more reasonable approach is to treat clean references as a set.

$$\gamma_p^{\text{clean}} = \{\gamma_{p,1}^{\text{clean}}, \dots, \gamma_{p,M}^{\text{clean}}\}$$

The noisy path is compared with all references in the set, and the minimum distance is taken as the deviation under the noise condition.

$$\Delta_p(\epsilon) = \min_{\gamma \in \gamma_p^{\text{clean}}} \frac{d_F(\gamma, \gamma_p^{\text{noisy}, \epsilon})}{\text{Len}(\gamma_p^{\text{clean}})}$$

For each experiment in a noisy setting, the noisy path extracted will match the "closest clean reference path". Then the median of the results will be taken as the robustness loss. It measures the typical geometric stability of the method against noise perturbations, taking into account clean ambiguity.

$$\text{RobustnessLoss}(p) = \text{median}_{\epsilon} [\Delta_p(\epsilon)]$$

To facilitate subsequent comparison with robustness on the same graph, this thesis further transforms the robustness loss into a robustness score that is considered higher is better:

$$\text{Robustness}(p) = \frac{1}{1 + \text{RobustnessLoss}(p)}$$

## 4 Results and Discussion

### 4.1 Clean setting overview

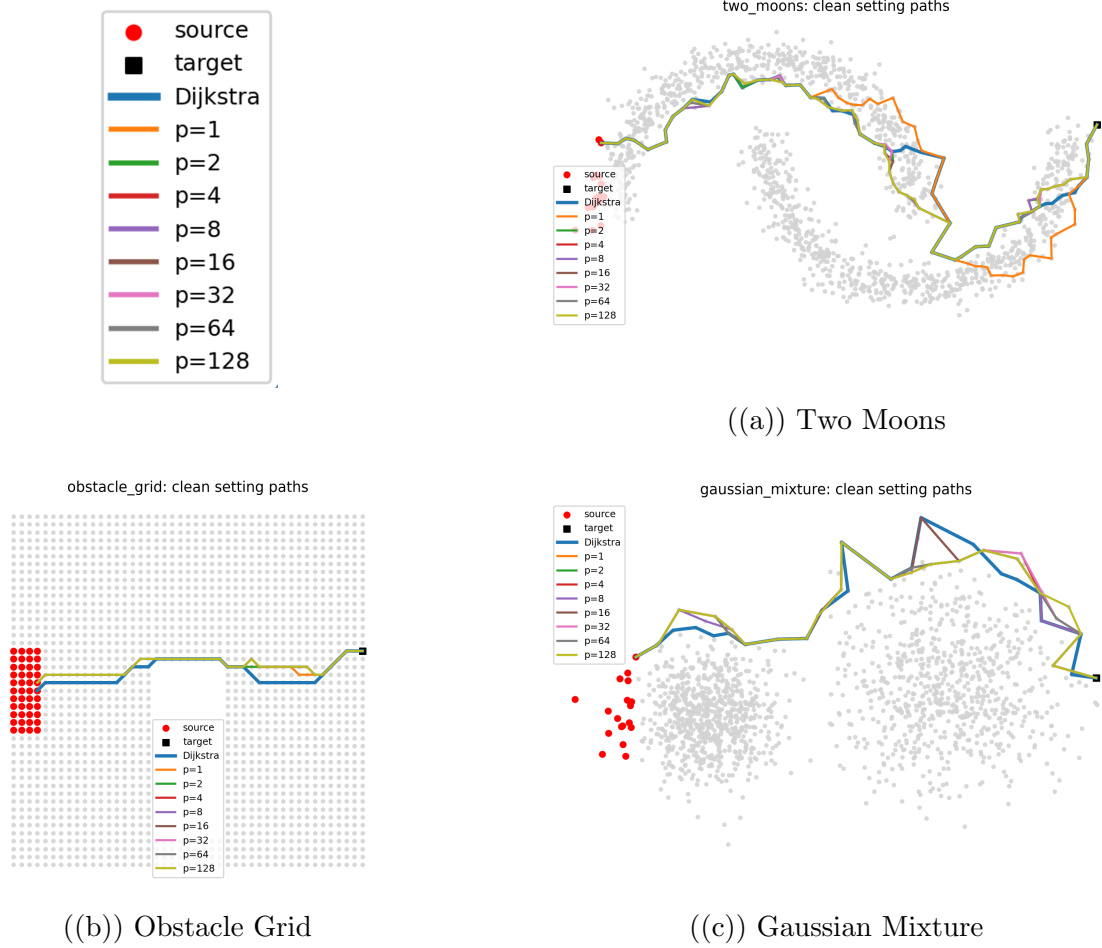


Figure 1: Clean-setting paths produced by Dijkstra and the  $p$ -eikonal method.

Figure 2: Clean-setting paths produced by Dijkstra and the  $p$ -eikonal method

Before analyzing the behavior of each method under noise perturbation, this thesis first compares the paths generated by Dijkstra's algorithm and the graph  $p$ -eikonal method of different values of  $p$  on clean datasets. Visual comparisons in Figure 2 show that the effect

of  $p$  is strongly dependent on the dataset.

The differences between paths are most significant for the "Two Moons" dataset. Although all methods generally follow a non-convex manifold structure, the paths generated by different  $p$  are not entirely consistent. Dijkstra paths and paths with smaller  $p$  tend to be closer to the outer arc, while some paths with larger  $p$  shift inward, closer to the gap between the two moons. This indicates that on non-convex manifold data, even in clean settings,  $p$  significantly influences the preferred propagation path.

For "Obstacle Grid", all methods choose the same global strategy: bypassing obstacles through upper-level paths. In this case, the differences between Dijkstra's path and the  $p$ -eikonal paths are mainly local: they appear in turning points and the piecewise linear shape of the paths.

For "Gaussian Mixture", the paths produced by Dijkstra's algorithm and the  $p$ -eikonal method are visually more similar to each other than in the previous two datasets. Most  $p$ -eikonal paths almost overlap with Dijkstra paths, with differences mainly concentrated in several transition vertices between the two clusters. Therefore, in the current implementation, the parameter  $p$  has a relatively limited impact on path geometry in this dataset.

Overall, the clean-setting comparison shows that the impact of  $p$  is not universal across datasets. It is strongest on Two Moons, moderate on Obstacle Grid, and relatively weak on Gaussian Mixture. For this reason, the clean-setting results are used in this thesis primarily as a baseline geometric comparison and as the reference for later noisy-path deviations, rather than as the main basis for parameter selection. The quantitative evaluation of robustness and fidelity is presented in the following sections.

## 4.2 Comparing how $p$ influences robustness under weighted noise

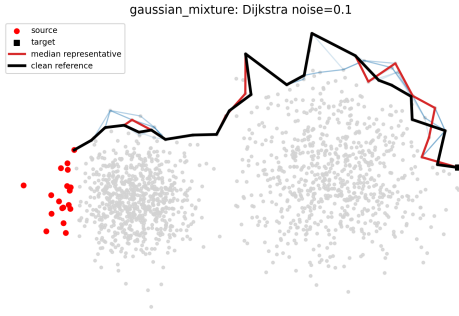
To further examine how the parameter  $p$  influences the robustness of the graph  $p$ -eikonal method, this subsection fixes the weighted noise level at

$$\delta \in \{0.05, 0.1, 0.2\}$$

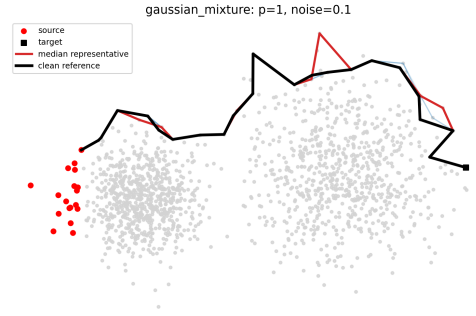
and compares the resulting path behavior for different values of  $p$  across the three datasets considered in this thesis: Gaussian Mixture, Obstacle Grid, and Two Moons. The thesis chooses  $p = 1$ ,  $p = 16$ , and  $p = 128$  as representative values as the low, middle, and high ends of the  $p$  range, in order to illustrate how the path solution evolves as  $p$  increases.

For each method and each dataset, 50 noisy realizations are generated under the same noise level, producing 50 corresponding noisy paths. Among these paths, the median representative path is selected to summarize the typical behavior under perturbation and is compared with the clean reference path. In the figures below, the black curve denotes the clean reference path, the red curve denotes the median representative path under noise, and the light blue curves indicate several noisy realizations. This visualization makes it possible to compare both the magnitude and the pattern of path variation directly across different values of  $p$ .

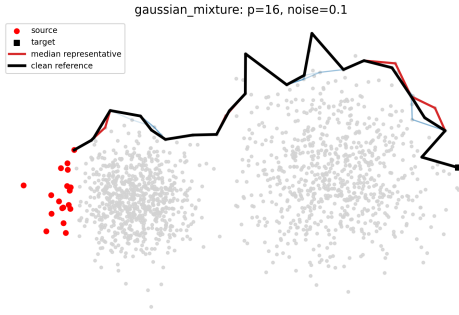
## Gaussian Mixture



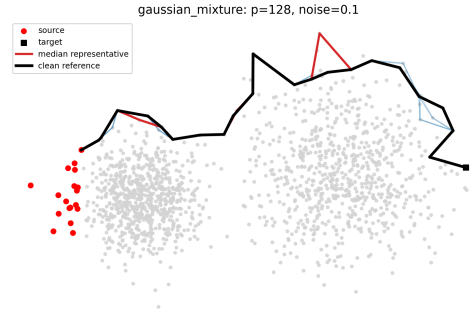
((a)) Dijkstra,  $\delta = 0.1$



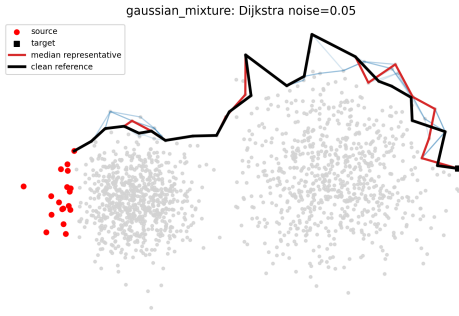
((b))  $p = 1$ ,  $\delta = 0.1$



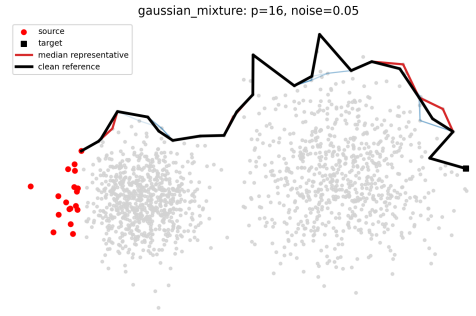
((c))  $p = 16$ ,  $\delta = 0.1$



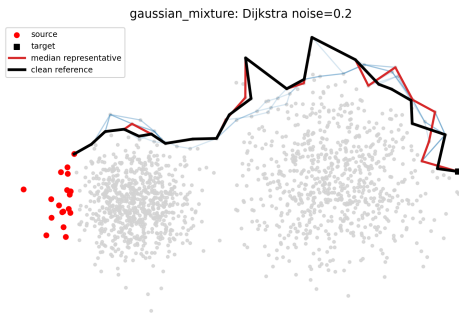
((d))  $p = 128$ ,  $\delta = 0.1$



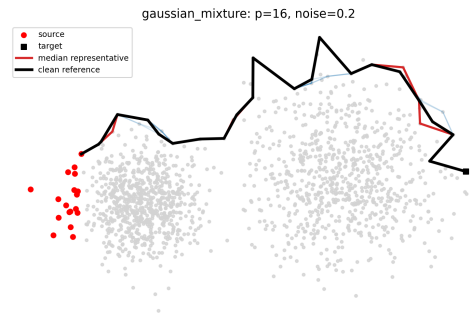
((e)) Dijkstra,  $\delta = 0.05$



((f))  $p = 16$ ,  $\delta = 0.05$

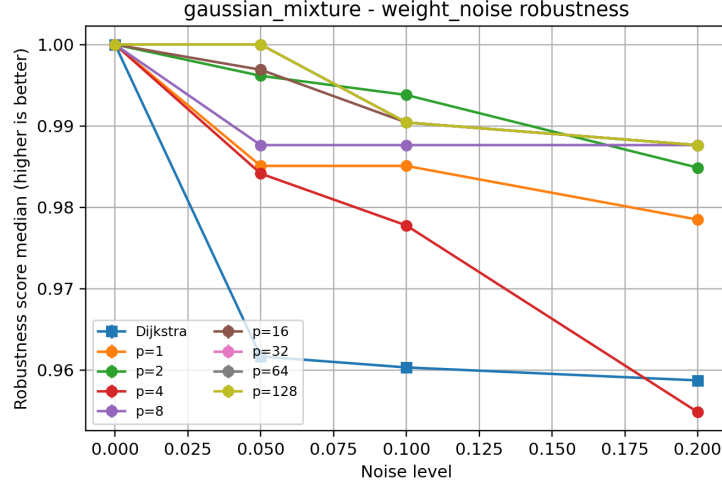


((g)) Dijkstra,  $\delta = 0.2$



((h))  $p = 16$ ,  $\delta = 0.2$

Figure 3: Representative noisy paths on Gaussian Mixture.



((a)) Robustness scores under different weighted noise levels

Figure 4: Robustness comparison on Gaussian Mixture under weighted perturbation.

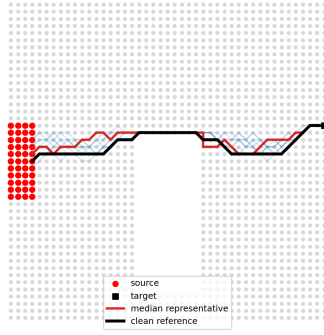
Figure 3 shows the results on the Gaussian Mixture dataset for Dijkstra’s algorithm and three representative  $p$ . The Gaussian Mixture dataset is the most sensitive of the three datasets, because the path must travel through a sparse region between two dense clusters, and perturbations in edge weights may alter which intermediate points are preferred.

From the visual comparison, Dijkstra’s median representative path deviates clearly from the clean reference. The noisy repeats also show visible branching, indicating that the discrete shortest path is sensitive to edge-weight perturbations. For  $p = 1$ , the graph  $p$ -eikonal method remains somewhat noise-sensitive, although its median path is already more stable than Dijkstra’s in some regions. As  $p$  increases, the median representative path lies much closer to the clean reference, and the noisy repeats cluster more tightly around it.

These observations agree with the robustness curves in Figure 4. Dijkstra’s robustness drops sharply once noise is introduced, whereas the  $p$ -eikonal solutions remain at a higher level. Moreover, the robustness scores show a clear ordering among the representative values, with  $p = 128$  performing better than  $p = 16$ , and  $p = 16$  performing better than  $p = 1$ .

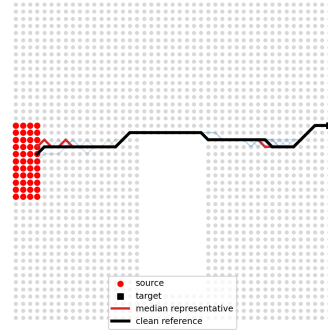
# Obstacle Grid

obstacle\_grid: Dijkstra noise=0.1



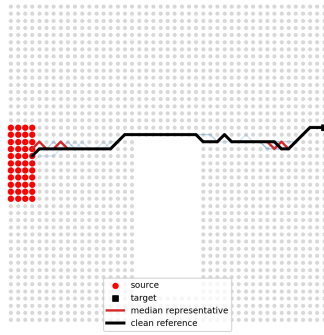
((a)) Dijkstra,  $\delta = 0.1$

obstacle\_grid: p=1, noise=0.1



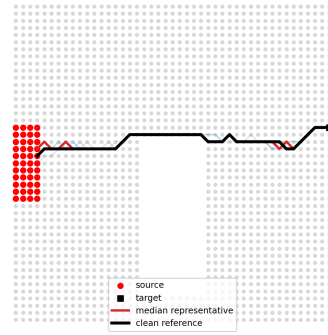
((b))  $p = 1$ ,  $\delta = 0.1$

obstacle\_grid: p=16, noise=0.1



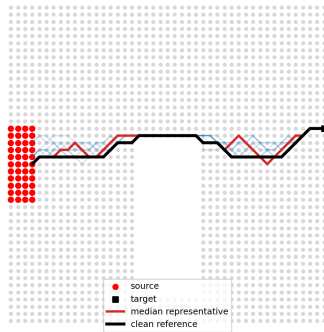
((c))  $p = 16$ ,  $\delta = 0.1$

obstacle\_grid: p=128, noise=0.1



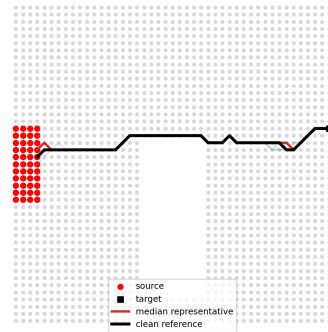
((d))  $p = 128$ ,  $\delta = 0.1$

obstacle\_grid: Dijkstra noise=0.05



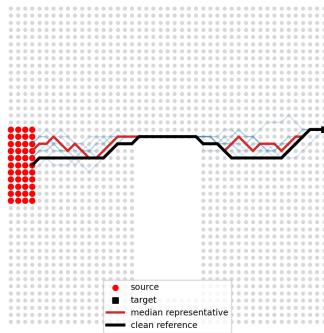
((e)) Dijkstra,  $\delta = 0.05$

obstacle\_grid: p=16, noise=0.05



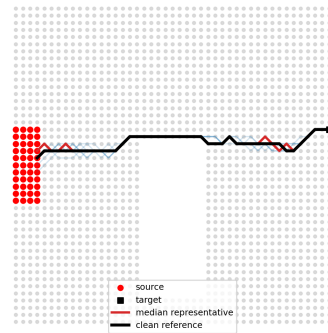
((f))  $p = 16$ ,  $\delta = 0.05$

obstacle\_grid: Dijkstra noise=0.2

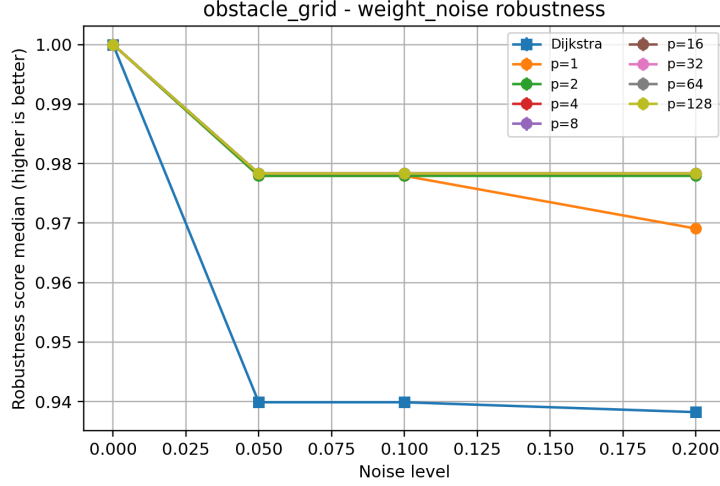


((g)) Dijkstra,  $\delta = 0.2$

obstacle\_grid: p=16, noise=0.2



((h))  $p = 16$ ,  $\delta = 0.2$



((a)) Robustness scores under different weighted noise levels

Figure 6: Robustness comparison on Obstacle Grid under weighted perturbation.

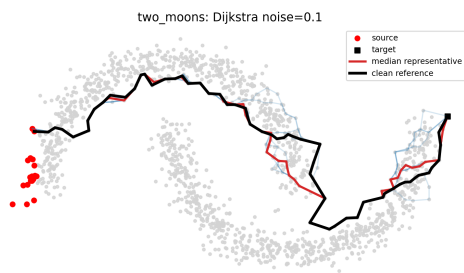
Figure 5 presents the corresponding paths on the Obstacle Grid dataset. Compared with Gaussian Mixture, the paths are more constrained geometrically: all methods must pass around the obstacle through an upper corridor. There isn't much room for global deviations.

Dijkstra's algorithm still shows visible noise sensitivity, especially in the local zig-zag behavior near the obstacle. However, the graph  $p$ -eikonal method is extremely stable for this dataset. At  $p = 1$ ,  $p = 16$  and  $p = 128$ , the results are nearly indistinguishable from the clean path: the median representative path almost overlaps with the clean reference, and the noisy repeats remain tightly concentrated.

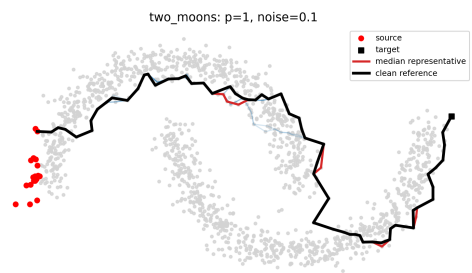
The robustness plot in Figure 6 confirms this visual impression. After the introduction of noise, Dijkstra drops to a noticeably lower robustness level, while all tested  $p$ -eikonal solutions remain clustered near the top of the plot. The differences among the tested values of  $p$  are very small, regardless of the choice of  $p$ , the method has already reached a stable regime. Therefore, for Obstacle Grid, the change of  $p$  barely influences robustness, and the

$p$ -eikonal family as a whole is substantially more stable than Dijkstra under weighted noise.

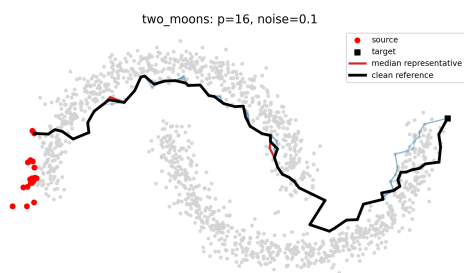
# Two Moons



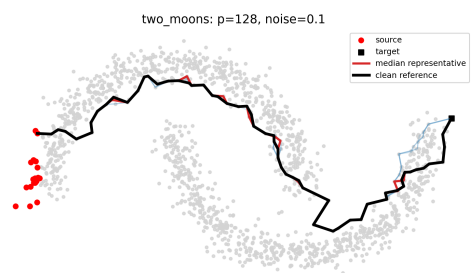
((a)) Dijkstra,  $\delta = 0.1$



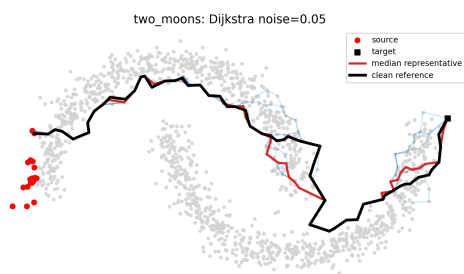
((b))  $p = 1$ ,  $\delta = 0.1$



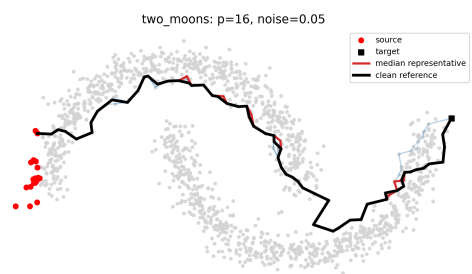
((c))  $p = 16$ ,  $\delta = 0.1$



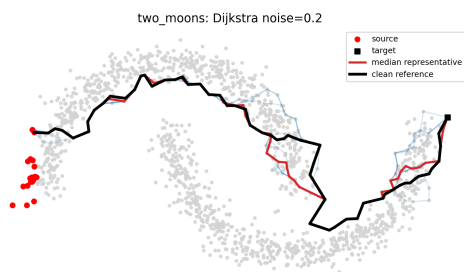
((d))  $p = 128$ ,  $\delta = 0.1$



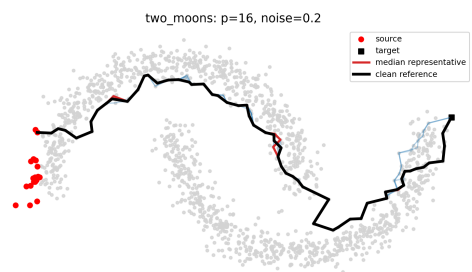
((e)) Dijkstra,  $\delta = 0.05$



((f))  $p = 16$ ,  $\delta = 0.05$

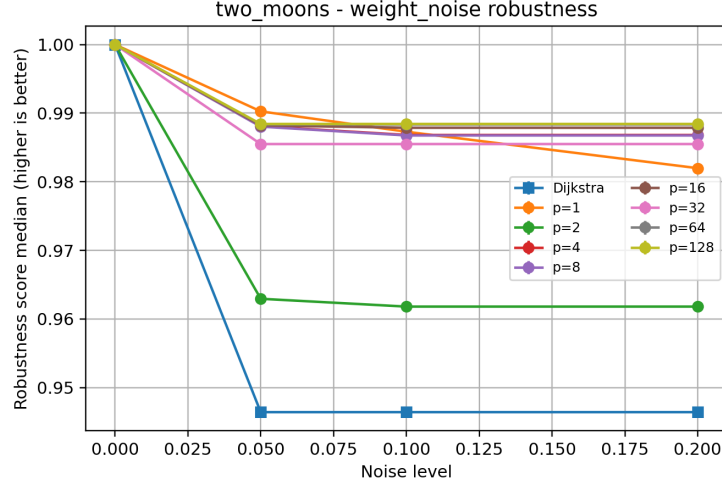


((g)) Dijkstra,  $\delta = 0.2$



((h))  $p = 16$ ,  $\delta = 0.2$

Figure 7: Representative noisy paths on Two Moons.



((a)) Robustness scores under different weighted noise levels

Figure 8: Robustness comparison on Two Moons under weighted perturbation.

Figure 7 shows the fixed-noise comparison on the Two Moons dataset. The path is constrained by the manifold structure of the moons, but it still has enough geometric freedom that perturbations can alter the preferred route, especially near the gap region and along the descending part of the left moon.

The visual results again show that Dijkstra’s algorithm is the most sensitive among the compared methods. Its median representative path deviates visibly from the clean reference, and some noisy repeats show substantial local variation. For  $p = 1$ , the median representative path is already closer to the clean reference than the Dijkstra path in several segments, although some deviations remain near the transition between the two moons. For  $p = 16$  and  $p = 128$ , the red and black paths nearly coincide over most of the trajectory, and the noisy repeats become more concentrated. This indicates that larger  $p$  provide better stability on the Two Moons dataset.

The robustness curves in Figure 8 support the same conclusion. Dijkstra’s algorithm shows the largest drop once noise is added. The  $p$ -eikonal method remains more stable for

nearly all tested values of  $p$ , and moderate-to-large values of  $p$  perform particularly well. Compared with Gaussian Mixture, the improvement here is more regular across  $p$ , although the differences among the larger values are still relatively small.

### 4.3 Fidelity–Robustness Trade-off and Parameter Selection

The previous section shows how  $p$  influences both path geometry in the clean setting and robustness under weighted perturbation. This subsection combines these two aspects by considering a fidelity–robustness trade-off. The goal is not to claim a universally optimal value of  $p$ , but to identify values that provide a reasonable balance between agreement with the clean reference and stability under noise.

#### Clean Fidelity

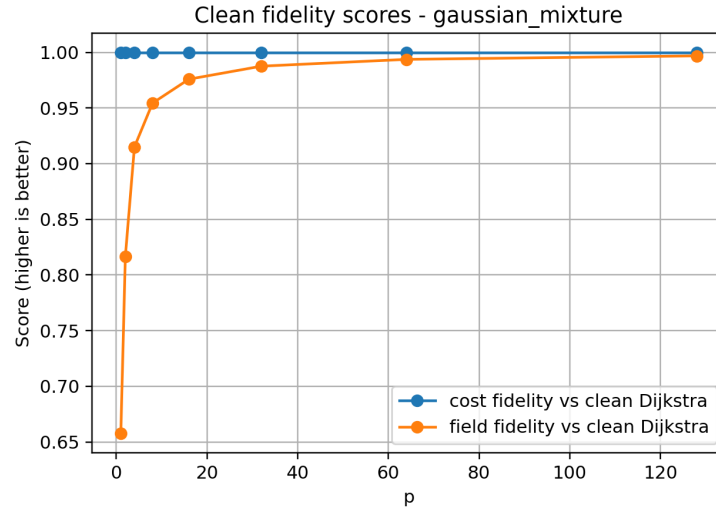


Figure 9: Clean fidelity scores on Gaussian

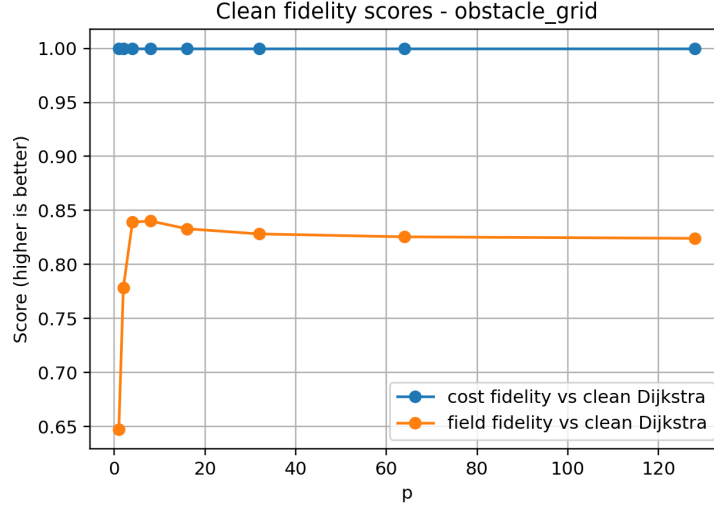


Figure 10: Clean fidelity scores on Obstacle Grid.

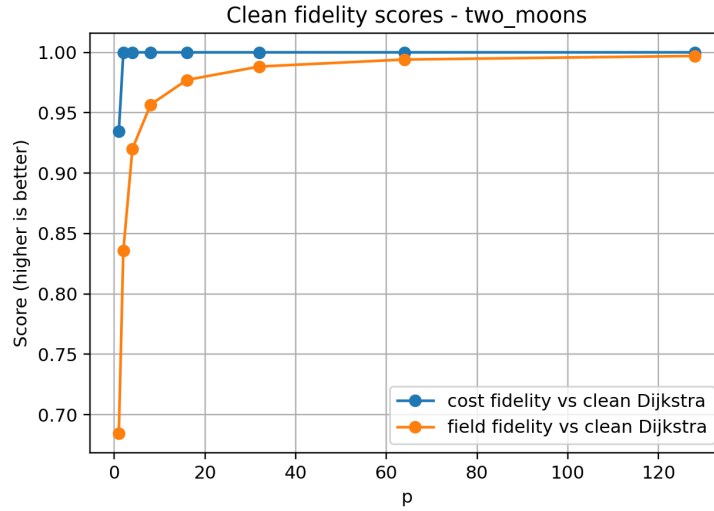


Figure 11: Clean fidelity scores on Two Moons.

Figures 10 and 11 summarize the clean fidelity scores for Obstacle Grid and Two Moons. Two fidelity measures are shown: a cost-based fidelity score and a field-based fidelity score, both measured relative to the clean Dijkstra reference. For both datasets, the cost fidelity quickly reaches 1 for a wide range of  $p$ , indicating that the clean path costs remain very close

to those of the Dijkstra reference. By contrast, the field fidelity is more discriminative. For small  $p$ , it is noticeably lower, and it increases as  $p$  becomes larger, eventually approaching a plateau. This indicates that although many  $p$ -eikonal solutions have nearly identical path cost to the clean Dijkstra path, the underlying value fields can still differ substantially when  $p$  is small.

## Fidelity–Robustness Curves

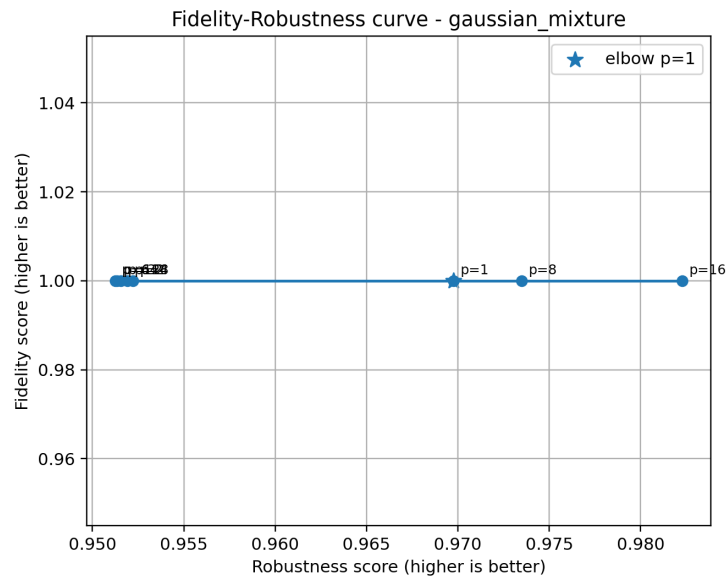


Figure 12: Fidelity–robustness curve on Gaussian Mixture.

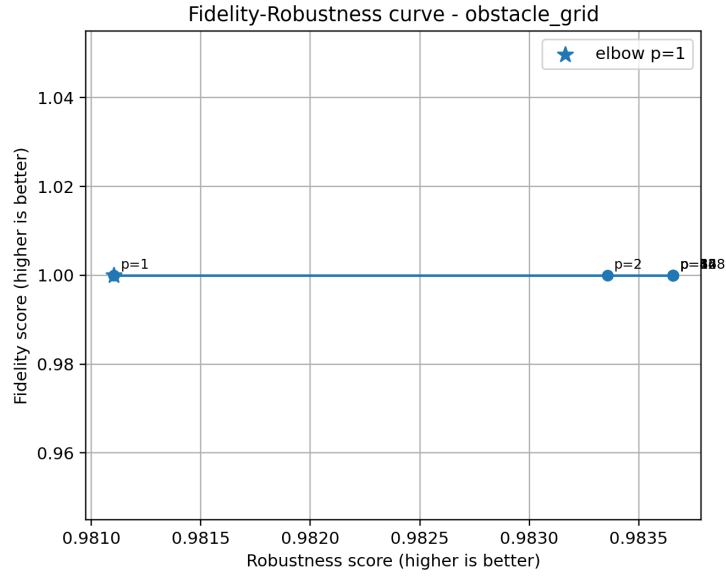


Figure 13: Fidelity-robustness curve on Obstacle Grid.

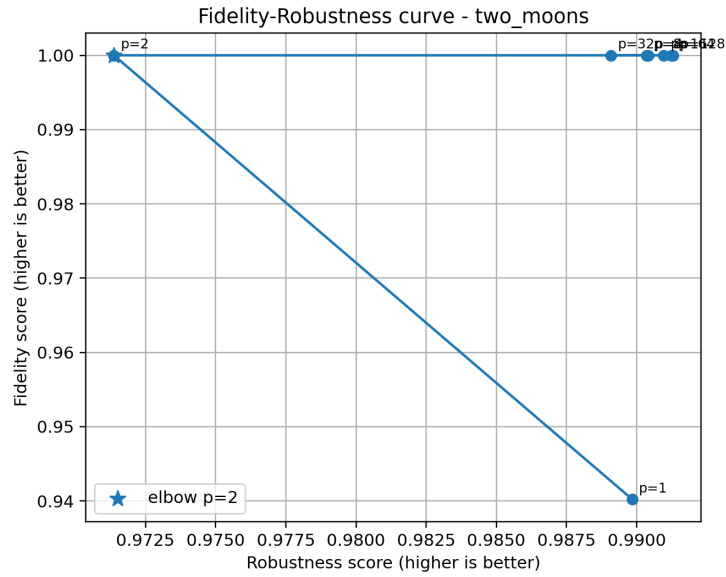


Figure 14: Fidelity-robustness curve on Two Moons.

To visualize the trade-off between the two criteria, Figures 12–14 plot each  $p$ -value as a point in the fidelity-robustness plane. Here the horizontal axis represents robustness under

weighted noise, while the vertical axis represents fidelity in the clean setting.

For Gaussian Mixture, the resulting curve is nearly horizontal because the fidelity values are essentially the same (which is 100%) across the tested  $p$ . The main distinction is therefore along the robustness axis. For Obstacle Grid, the situations are almost the same: fidelity is almost constant at the top of the plot, while robustness varies only slightly. For Two Moons, however, the trade-off is more informative. In this dataset,  $p = 1$  has high robustness but noticeably lower fidelity than the remaining values, whereas  $p \geq 2$  yields both very high fidelity and robust behavior.

Overall, these curves suggest that identifying a single optimal value of  $p$  is not especially meaningful in the present experiments, since many tested values already perform very well. Rather than isolating one uniquely best parameter, the fidelity–robustness plots indicate that there is a broad range of acceptable  $p$ -values for which both fidelity and robustness remain high. Thus, in this thesis, the main role of the fidelity–robustness analysis is not to determine a precise optimum, but to show that the  $p$ -eikonal method performs stably across a substantial parameter range.

## 4.4 Discussions on Limitations

The results in this chapter show that dataset geometry still affects the behavior of the graph  $p$ -eikonal method. On Obstacle Grid, the obstacle structure tightly constrains the admissible paths, so different  $p$ -values behave very similarly once one moves away from Dijkstra. On Two Moons, the manifold geometry allows slightly more variation, and small  $p$ -values may differ from moderate ones. Gaussian Mixture is the most flexible case, since the path must cross a sparse region between clusters, making local perturbations more influential. However, although these structural differences produce some variation across datasets, the overall differences among most tested  $p$ -values are not large. For this reason, selecting a single

optimal  $p$  is not especially meaningful in the present experiments; in practice, a broad range of moderate or large  $p$ -values already performs well.

There are also several limitations. First, the experiments consider only weighted edge perturbations, and other noise types, such as abnormal shortcut edges, may lead to different conclusions. Second, the fidelity metrics are defined relative to the clean Dijkstra reference, so they measure agreement with that baseline rather than absolute correctness. Third, the analysis is based on a small number of synthetic datasets, which may not fully represent more complex real-world graphs. These limitations suggest that the conclusions should be interpreted as comparative rather than universal.

## 5 Test on MNIST

To further examine whether the observations from the synthetic datasets remain valid in a higher-dimensional classification setting, the thesis tests the graph-based Dijkstra and  $p$ -eikonal methods on a subset of the MNIST handwritten digit dataset. The purpose of this experiment is not to compete with highly optimized deep learning classifiers, but to investigate whether the same qualitative conclusions observed in Chapter 4 also hold when the methods are used as distance-based classifiers on a real benchmark dataset.

In this experiment, the data are first represented in a reduced feature space, and a  $k$ -nearest-neighbor graph is constructed. For each digit class, a small labeled subset is treated as the source set, and classification is performed by comparing the propagated distances from different class sources. The predicted label of each test sample is determined by the class with the smallest propagated distance. This produces a graph-based semi-supervised classification framework that is consistent with the distance-propagation viewpoint used throughout this thesis.

## 5.1 Accuracy and Macro-F1 Across Different $p$

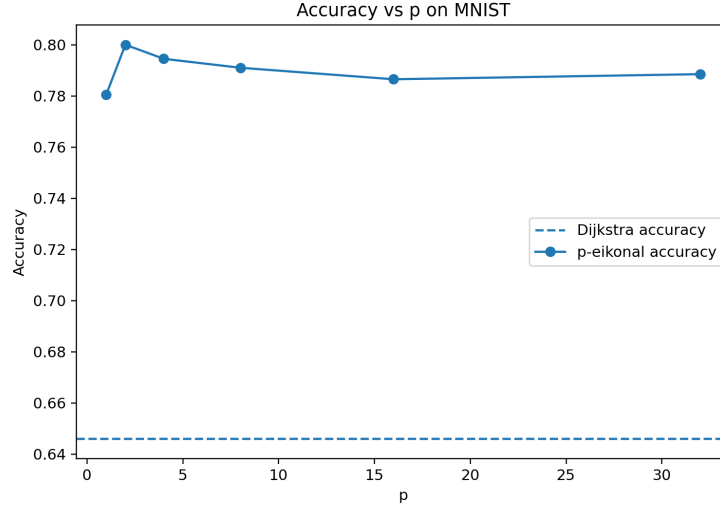


Figure 15: Classification accuracy versus  $p$  on MNIST.

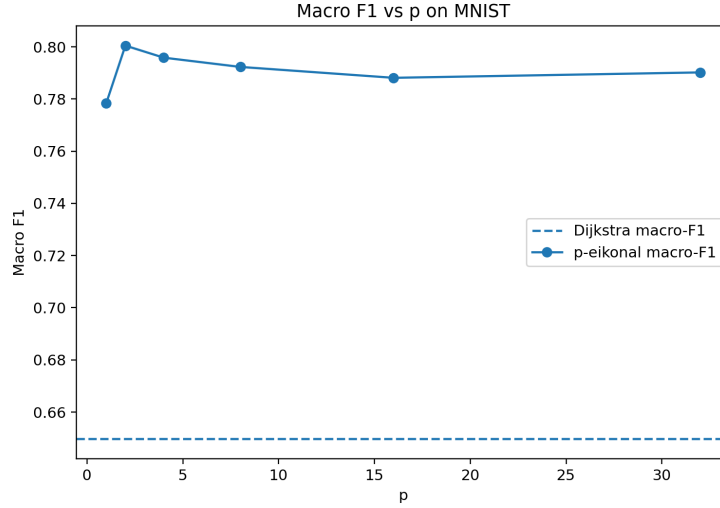


Figure 16: Macro-F1 score versus  $p$  on MNIST.

Figure 15 shows the classification accuracy of the  $p$ -eikonal method for different values of  $p$ , together with the Dijkstra baseline. Figure 16 reports the corresponding macro-F1 scores. The numerical results show a clear improvement of the  $p$ -eikonal method over Dijkstra's

algorithm. The baseline of Dijkstra’s algorithm attains an accuracy of 0.646 and a macro-F1 score of 0.6497, whereas all tested  $p$ -eikonal models substantially outperform it. The best result is achieved at  $p = 2$ , with accuracy 0.8000 and macro-F1 0.8005.

These results indicate conclusions consistent with those in Chapter 4: the parameter  $p$  does not matter much, since the differences among  $p$  become relatively small. In particular, the MNIST experiment again suggests that a small or moderate value of  $p$  is often sufficient, rather than a very large one.

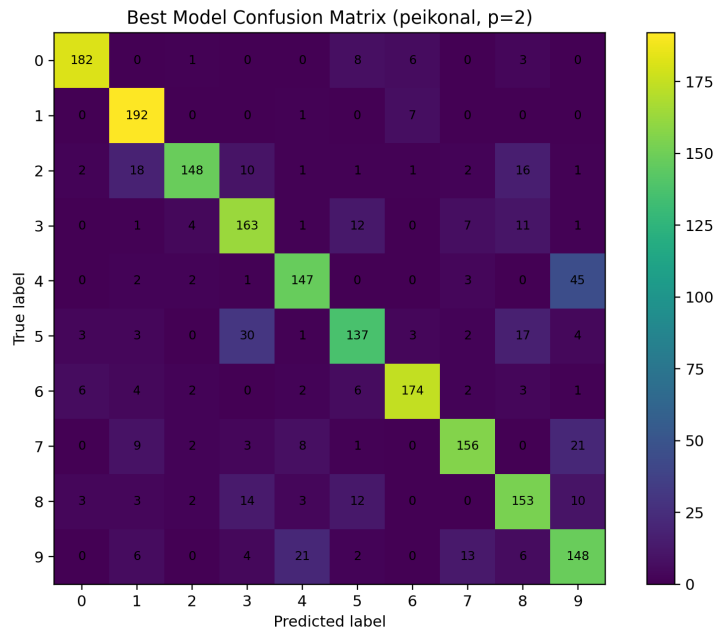


Figure 17: Confusion matrix of the best MNIST model ( $p = 2$ ).

Figure 17 shows the confusion matrix of the best-performing model, namely the  $p$ -eikonal classifier with  $p = 2$ . Several classes are recognized quite reliably. However, some confusions remain visible. In particular, digit 4 is frequently misclassified as 9, and digit 5 is often confused with 3 and 8. These errors are natural in a graph-distance framework, since visually similar digits may form neighboring structures in the feature graph, making their propagated distance fields overlap.

## 6 Conclusion

This thesis presented a comparative study of Dijkstra-type shortest-path methods and graph  $p$ -eikonal methods on graphs. The main goal was to understand how these two approaches differ in terms of path geometry, agreement with clean reference solutions, robustness under perturbations, and practical behavior on both synthetic and real datasets.

The experimental results show that the two methods are closely related but not equivalent. On the synthetic datasets, Dijkstra tends to emphasize discrete shortest-path behavior, while the  $p$ -eikonal method produces a family of distance propagations whose behavior depends on the parameter  $p$ . As  $p$  varies, the computed solutions may change in both geometric shape and robustness. In particular, the results indicate that the  $p$ -eikonal method can provide smoother and more stable distance fields than Dijkstra, especially in settings where the graph geometry is affected by weighted perturbations. This supports the view that the  $p$ -eikonal method is not merely another implementation of shortest-path search, but a more flexible framework for distance construction on graphs.

The clean-setting experiments further show that fidelity and robustness should be viewed together, but they do not support a strong need to identify a single optimal value of  $p$ . On some datasets, the fidelity metric is somewhat sensitive to  $p$ , reflecting structural differences in how the methods propagate through the graph. On others, however, the fidelity–robustness curve is nearly degenerate, and many nearby  $p$ -values perform similarly well. Taken together, these results suggest that the choice of  $p$  is not a decisive issue in the present experiments: once  $p$  leaves the least favorable small- $p$  regime, a broad range of values already gives high fidelity and strong robustness.

The MNIST experiment leads to the same overall conclusion in a higher-dimensional classification setting. When graph-based distance propagation is used for classification, the  $p$ -eikonal method clearly outperforms the Dijkstra baseline, showing that it provides a more

informative class-wise distance field. At the same time, although the best result is attained at a small or moderate value of  $p$ , the performance differences among the tested  $p$ -values are limited. This is consistent with the synthetic datasets: changing  $p$  can produce some variation, but in practice many choices already perform well, so selecting an exact optimum is of limited importance.

Overall, the main conclusion of this thesis is that the  $p$ -eikonal method offers a meaningful advantage over the Dijkstra method when one is interested not only in a single shortest path, but in a distance propagation that better reflects the geometry of the data and remains stable under perturbation. However, this advantage should not be understood as evidence for a universally optimal parameter choice. Rather, the practical value of the method lies in providing a flexible family of graph-based distance models for which the performance is relatively stable across a broad range of  $p$ -values.

There are several possible directions for future work. First, more systematic experiments could be conducted on additional real datasets and graph constructions to test whether the same conclusions remain valid beyond the datasets considered here. Second, the parameter-selection rule could be improved by developing more principled criteria for identifying stable operating regions in the fidelity–robustness plane. Third, it would be valuable to investigate stronger feature representations and larger-scale graph constructions in classification tasks, in order to better understand how the performance of  $p$ -eikonal methods depends on the underlying data representation. Finally, further theoretical work could help clarify the relationship between graph-based  $p$ -eikonal distances, continuum Hamilton–Jacobi models, and robustness properties under different types of perturbations.

In summary, this thesis shows that comparing Dijkstra and  $p$ -eikonal methods is not simply a question of which one finds a shorter path. Rather, it reveals a broader distinction between discrete shortest-path optimization and Hamilton–Jacobi-type distance propaga-

tion. From this perspective, the  $p$ -eikonal framework provides a useful and interpretable extension of graph-based distance methods, with clear advantages in robustness and flexibility, provided that the parameter  $p$  is chosen in an appropriate range.

## 7 References

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